Total No. of Printed Pages: 2

SUBJECT CODE NO: - RNP-8003
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Mechanical) (NEP) Sem - III
Examination November / December 2025 (To be held in February-2026)
Engineering Thermodynamics

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Q.no 2, 3, 4, 5, 6 and 7

Section A

Q1 **Solve any Ten** **20**

a) What is a steady flow process?

b) A machine that violates the first law of thermodynamics is known as --------

c) State the Kelvin-Planck statement of the second law of thermodynamics.

d) The device that transfers heat from a low-temperature body to a high-temperature body is called a ---------

e) The area under a process curve on a T-S diagram represents -----------

f) Match the Pair

A	B
(a) Carnot Cycle	(i) Constant pressure heat addition
(b) Otto Cycle	(ii) Constant volume heat addition
(c) Diesel Cycle	(iii) Reversible cycle with maximum efficiency
(d) Brayton Cycle	(iv) Gas turbine cycle

g) The cycle used in petrol engines is called the -------- cycle.

h) What is meant by "dead state"?

i) What is the dryness fraction of steam?

j) Define calorific value of a fuel.

k) The point at which all three phases of water coexist is called the -----------.

l) The apparatus used for flue gas analysis is known as ------------.

Section B

Q2	a)	What are the limitations of the first law of thermodynamics? Explain with examples.	**03**
	b)	A steam turbine operates under steady flow condition receiving steam at the following stage, Pressure 15 bar internal energy 2700 Kj/Kg, specific volume 0.17 m^3/ kg and velocity 100 m/s. The exhaust of steam from the turbine is at 0.1 bar with internal energy 2175 Kj/kg, specific volume 15 m^3/kg And velocity 300 m/s. the turbine developes 35 KW and heat loss over the surface of turbine is 20 Kj/kg. Determine the steam flow rate through the turbine?	**07**
Q3	a)	State and explain the Kelvin-Planck and Clausius statements of the second law.	**05**
	b)	A reversed Carnot cycle operates as refrigerator has capacity of 100 kJ/s while operating between temperature limits of -20°c & 35°C Determine power input & COP.	**05**
Q4	a)	Explain the Clausius inequality and state its significance.	**05**
	b)	Define availability and irreversibility. Explain how they are related to energy degradation.	**05**
Q5	a)	Derive the air standard efficiency of the Diesel cycle	**05**
	b)	Draw and explain the T-S and P-V diagrams for the Brayton cycle and derive its efficiency formula	**05**
Q6	a)	Define dryness fraction and explain how it can be measured by a throttling calorimeter.	**05**
	b)	A pressure cooker contains 2kg of dry & saturated steam at 5bar. Find the quantity of heat that must be rejected so as to reduce quality upto 60% dry. Also find pressure & temp at new state.	**05**
Q7	a)	Define calorific value. Explain the difference between higher and lower calorific values.	**05**
	b)	A fuel contains 82% C, 16% H2, and 2% O2 Calculate the theoretical air-fuel ratio.	**05**